

CENTRALIZED PRISON MANAGEMENT SYSTEM

Product Requirements Document | v1.0

Project Centralized Prison Management System (CPMS)	Version 1.0 — Initial Release
Status Draft	Date April 2026
Team Mohammed · Youssef · Pavly · Menna · Malak	

1. Product Overview

1.1 Goals & Objectives

1.2 Scope

2. System Structure

2.1 Physical Hierarchy

2.2 Prison Data Model

2.2.1 Core Facility Information

2.2.2 Blocks and Cells Information

2.2.3 Super Admin

2.2.4 Officers

2.2.5 Facility Features (Boolean Flags)

3. User Roles & Permissions

3.1 Super Admin

3.2 Prison Manager (Prison Admin)

3.3 Officer

3.4 Public User (Visit Request Portal)

3.5 Shift Management

3.5.1 Shift Assignment Rules

3.5.2 Shift Assignment Data Model

4. Inmate Management

4.1 Inmate Placement Workflow

4.1.1 Placement Criteria (Super Admin)

4.2 Inmate Data Model

4.2.1 Personal Information

4.2.2 Legal Case Information

4.3 Transfers

5. Visit Management

5.1 Public Visit Request Portal

5.2 Visitor Data Model

5.2 Visit Data Model

5.3 Legal Visits

6. Incident & Disciplinary Management

6.1 Incident Reports

6.2 Disciplinary Log

7. Health Care

7.1 Doctors

7.2 Visits Reporting

8. Machine Learning Features

8.1 Risk Behavior Prediction

8.2 Overcrowding Prediction

8.3 Recidivism Risk Scoring

9. Dashboard Specifications

9.1 Super Admin Dashboard

9.2 Prison Manager Dashboard

9.3 Officer Dashboard

10. Non-Functional Requirements

10.1 Security

10.2 Performance

10.3 Availability

10.4 Scalability

10.5 Compliance

11. Glossary

1. Product Overview

The Centralized Prison Management System (CPMS) is a unified digital platform designed to manage all aspects of prison administration across multiple correctional facilities. The system consolidates inmate records, staff management, security operations, inter-prison transfers, visit scheduling, and incident reporting into a single, role-based platform — with AI-powered predictive analytics layered on top.

Attribute	Detail
Product Name	Centralized Prison Management System (CPMS)
Primary Users	Super Admin, Prison Managers, Officers, Public (Visit Requests)
Architecture	Multi-prison, multi-role, centralized database
AI Features	Risk Prediction, Overcrowding Forecast, Recidivism Scoring
Access Model	Role-Based Access Control (RBAC)

1.1 Goals & Objectives

- Centralize management of all prisons under a unified platform accessible by authorized roles.
- Eliminate paper-based inmate, incident, and visit records with a fully digital workflow.
- Enable data-driven decision-making through real-time dashboards and ML predictions.
- Streamline inter-prison transfer requests, approvals, and inmate placement.
- Provide a public-facing portal for submitting and tracking visit requests.
- Ensure accountability through full audit logs of incidents, movements, and disciplinary actions.

1.2 Scope

In scope:

- Multi-prison management with hierarchical structure (Prison → Block → Cell)
- Three primary internal roles: Super Admin, Prison Manager, Officer
- Public-facing Visit Request portal
- Inmate lifecycle management (admission → placement → transfer → release)
- Incident and disciplinary record keeping
- ML-based predictive features (risk behavior, overcrowding, recidivism)

Out of scope (Phase 1):

- Financial/payroll management for staff

- Parole board integrations
- Biometric access control hardware integration

2. System Structure

2.1 Physical Hierarchy

The system models the physical layout of the correctional estate as a three-level hierarchy:

Level	Entity	Description
Level 1	Prison / Facility	A standalone correctional institution with its own staff, blocks, and capacity.
Level 2	Block	A named wing or section within a prison (e.g., Block A — Maximum Security).
Level 3	Cell	An individual cell within a block with defined capacity and occupancy.

2.2 Prison Data Model

Each prison record stores the following attributes:

2.2.1 Core Facility Information

Field	Type	Description
Prison ID	Unique Identifier	Auto-generated unique ID for each prison.
Manager	Foreign Key	The officer who Manage
Name	String	Official name of the facility.
Location	Enum	City/Governorate where the facility is situated.
Type	Enum	e.g., Maximum Security, Minimum Security, Remand, Juvenile, Women's.
Security Level	Enum	
Total Capacity	Integer	Maximum number of inmates the facility can hold.
Current Occupancy	Integer	Number of currently housed inmates (auto-computed).

2.2.2 Blocks and Cells Information

Field	Type	Description
Block ID	Unique Identifier	Auto-generated unique ID for each block.
Prison ID	Foreign Key	
Capacity	Integer	
Current occupancy	Integer	
Security level	Enum	
Number of cells	Integer	

Field	Type	Description
Cell ID	Unique Identifier	Auto-generated unique ID for each prison.
Block ID	Foreign Key	Official name of the facility.
Prison Id	Foreign Key	
Capacity	Integer	
Current occupancy	Integer	

2.2.3 Super Admin

Field	Type	Description
National ID	Unique Identifier	
Name	String	
Phone	String	
Address	String	
Email	String	
Password	Password	

2.2.4 Officers

Field	Type	Description
National ID	Unique Identifier	
Prison ID	Foreign Key	
Name	String	
Address	String	
Phone	String	
Email	String	
Password	Password	

2.2.5 Facility Features (Boolean Flags)

Feature	Description
Infirmary / Hospital	Does the facility have a resident surgical or medical unit?
Workshops	Are there vocational training areas (carpentry, textiles, etc.)?
Agricultural Ward	Does the prison engage in farming or agricultural activities?
Visitation Hall	Does the facility have a dedicated family visitation hall, and what is its capacity?

3. User Roles & Permissions

CPMS uses Role-Based Access Control (RBAC). Every user belongs to exactly one role, and each role has a strictly defined set of permissions and a dedicated dashboard.

3.1 Super Admin

Super Admin	
Description The central authority who oversees all correctional facilities nationwide. Has read/write access to all prisons, all inmate records, and all system-level configurations.	Dashboard KPIs • Total prisoners per prison • Occupancy rate across all prisons • Transfer statistics (pending / approved / denied) • System-wide alerts (overcrowding, high-risk inmates)
Key Features & Permissions • Add, edit, and deactivate prison facilities • Assign inmates to specific prisons based on placement criteria • View real-time occupancy data for any prison, block, or cell • View, approve, and reject inter-prison transfer requests • View all inmate records, legal case files, and health data • Access all ML prediction outputs • Manage system-level user accounts and role assignments	

3.2 Prison Manager (Prison Admin)

Prison Manager	
Description Manages the day-to-day operations of a single assigned prison. Has full visibility into inmates, staff, occupancy, and scheduling within their facility.	Dashboard KPIs • Occupancy rate by block and by cell • Active incidents count • Pending / approved visit requests • Upcoming releases (next 30 days) • Officer shift assignments
Key Features & Permissions • View all inmate profiles and legal case data for their prison • Access all ML prediction outputs • Assign inmates to specific blocks and cells • View all officers and doctors assigned to their prison • Add time Slots for the visits • Accept or reject public visit requests • View occupancy at block and cell level • Submit inter-prison transfer requests for inmates • View disciplinary logs and incident reports • Manage internal movement logs	

3.3 Officer

Officer	
Description A guard or security officer assigned to one or more blocks within a prison. Responsible for reporting incidents and monitoring the cells under their supervision.	Dashboard KPIs - Occupancy of cells in their assigned block(s) - Recent incidents in their block - Today's scheduled movements - Active solitary confinement records
Key Features & Permissions - Report incidents: fights, self-harm, escape attempts - Log disciplinary actions and violation records - Record solitary confinement orders (up to 30 days) - Log internal movements (cell transfers) and external movements - View inmate profiles for inmates in their assigned block - Cannot approve visits or submit transfer requests	

3.4 Public User (Visit Request Portal)

Any member of the public can access the Visit Request portal without creating an account. They submit a visit request by:

- Entering the unique inmate ID of the person they wish to visit
- Providing their own National ID, Full Name, relationship to the inmate, and contact information
- Selecting from available pre-defined time slots

The prison manager reviews and approves or denies the request. The visitor is notified of the decision.

3.5 Shift Management

The Prison Manager is responsible for assigning officer shifts to blocks within their facility. Shifts are the primary mechanism by which blocks are staffed and monitored at all times.

Shift	Hours	Description
Morning Shift	06:00 – 14:00	Day-time operations, inmate activities, visits, and court movement scheduling.
Afternoon Shift	14:00 – 22:00	Evening supervision, incident monitoring, and lockdown procedures.
Night Shift	22:00 – 06:00	Overnight watch; reduced activity, emergency response readiness.

3.5.1 Shift Assignment Rules

- Each block must have at least one officer assigned per shift at all times.
- An officer can only be assigned to one block per shift slot.
- The Prison Manager assigns shifts via the Shift Management page, selecting officer, block, shift type, and date range.
- Officers can view their own upcoming shift schedule from their dashboard.
- Shift history is logged and retained for audit purposes.

3.5.2 Shift Assignment Data Model

Field	Type	Description
Shift ID	Auto-generated	Unique identifier for each shift assignment.
Officer	Foreign Key	The officer assigned to the shift.
Block	Foreign Key	The block the officer is responsible for during this shift.
Shift Type	Enum	Morning / Afternoon / Night.
Date	Date	The calendar date of the shift.
Start Time	Time	Shift start time (auto-filled from shift type).
End Time	Time	Shift end time (auto-filled from shift type).

4. Inmate Management

4.1 Inmate Placement Workflow

The placement process follows a two-stage workflow:

Stage	Actor	Action
1. Prison Assignment	Super Admin	Assigns a newly admitted inmate to a specific prison based on placement criteria below.
2. Cell Placement	Prison Manager	Once the inmate arrives, the Prison Manager assigns them to a specific block and cell within the prison.

4.1.1 Placement Criteria (Super Admin)

The Super Admin evaluates the following factors when selecting a prison:

- Location — proximity to the inmate's court jurisdiction or family region
- Security Level — matching inmate risk profile to facility type (e.g., Maximum Security)
- Legal Case & Sentence — nature of offense and sentencing requirements
- Sentence Duration — long-term inmates may require specific facilities
- Age — juvenile inmates must be placed in appropriate facilities
- Health Conditions — inmates requiring medical care must be placed in prisons with infirmaries

4.2 Inmate Data Model

4.2.1 Personal Information

Field	Type	Notes
Unique Inmate ID	Auto-generated	System-generated identifier for internal use.
Full Name	String	As per National ID document.
Date of Birth	Date	Used for age calculation and juvenile classification.
Gender	Enum	Male / Female / Other.
Nationality	String	Country of citizenship.
Occupation	String	Civilian occupation prior to incarceration.
National ID	String	Government-issued national identification number.
Education level	Enum	
Start Date	Date	Date of admission into the prison system.
Expected Release Date	Date	Calculated from sentence start date + duration.

Assigned Prison	Foreign Key	Links to the prison entity.
Assigned Block	Foreign Key	Links to the block entity within the prison.
Assigned Cell	Foreign Key	Links to the specific cell within the block.

4.2.2 Legal Case Information

Field	Type	Notes
Case Number	String	Official court case reference number.
Crime Type	Enum	
Court Name	String	Name of the presiding court.
Sentence Duration	Duration	Total sentence length (e.g., 5 years, life).
Inmate ID	Foreign Key	Link to the inmate

4.3 Transfers

An inter-prison transfer can be requested by a Prison Manager for any inmate within their facility. The request is reviewed and approved or denied by the Super Admin.

Field	Type	Description
Transfer ID	Auto-generated	
Requesting Prison	Foreign Key	The prison initiating the transfer request.
Destination Prison	Foreign Key	The proposed receiving facility.
Reason for Transfer	String	Clinical need, security reclassification, overcrowding, court order, etc.
Inmate ID	Foreign Key	The inmate subject to transfer.
Status	Enum	Pending / Approved / Denied.
Approval Date	Date	Date the Super Admin acted on the request.

5. Visit Management

5.1 Public Visit Request Portal

A publicly accessible page allows anyone to request a visit to an inmate. No account creation is required.

Required inputs from the visitor:

- Inmate National ID (to identify the correct inmate)
- Visitor's Full Name
- Visitor's National ID
- Relationship to Inmate (e.g., spouse, parent, sibling, friend, legal counsel)
- Contact information (phone and email)
- Selection from available time slots

5.2 Visitor Data Model

Field	Type	Description
Visit ID	Foreign Key	Unique visit reference.
National ID	String	Linked inmate record.
Full Name	String	Name of the person visiting.
Relationship	Enum	Spouse, Parent, Sibling, Friend, Lawyer, Other.
Phone	String	Phone number
Email	String	Email

5.2 Visit Data Model

Field	Type	Description
Visit ID	Auto-generated	Unique visit reference.
Inmate National ID	Foreign Key	Linked inmate record.
Date	Date	Scheduled date of visit.
Time Slot	Enum	Selected from pre-defined available slots.
Duration	Integer	Duration in minutes (e.g., 30, 60).
Status	Enum	Pending / Approved / Denied.
Denial Reason	String	If denied, reason provided to visitor.
Visit Type	Enum	Regular / Legal (lawyer consultation).

5.3 Legal Visits

Legal visits (lawyer–client consultations) are tracked separately from regular family visits to ensure attorney–client privilege. Legal visits:

- Are flagged with Visit Type = Legal
- Are not subject to the same scheduling restrictions as regular visits

6. Incident & Disciplinary Management

6.1 Incident Reports

Officers are responsible for reporting any security incidents that occur within their assigned blocks. Each report captures:

Field	Type	Description
Incident ID	Auto-generated	Auto-generated unique reference.
Type	Enum	Fight, Self-Harm, Escape Attempt, Property Damage, Assault on Staff, Other.
Date & Time	Time	Exact timestamp of the incident.
Location	Foreign Key	Prison, Block, and Cell where the incident occurred.
Inmates Involved	Foreign Key	Linked inmate IDs of all parties involved.
Reporting Officer	Foreign Key	The officer who filed the report.
Staff Involved	Foreign Key	Other officers or staff present during the incident.
Description	String	Narrative description of the incident.
Action Taken	String	Immediate response measures applied.

6.2 Disciplinary Log

All punishments administered to inmates must be recorded in the Disciplinary Log. This is a mandatory registry and cannot be deleted.

Field	Type	Description
Inmate ID	Foreign Key	The inmate receiving the disciplinary action.
Incident ID	Foreign Key	Reference to the triggering incident report.
Punishment Type	Enum	Loss of privileges, solitary confinement, transfer to high-security room, etc.
Solitary Confinement Duration	Integer	If applicable: duration in days (maximum 30 days as per regulations).
Imposed By	Foreign Key	The officer or manager who issued the punishment.
Date Imposed	Date	Date the punishment took effect.
End Date	Date	Date the punishment expires.
Notes	String	Any additional details or observations.

7. Health Care

7.1 Doctors

Field	Type	Description
National ID	Auto-generated	Auto-generated unique reference.
Name	Foreign Key	
address	String	
Phone	String	
Prison ID	Foreign Key	

7.2 Visits Reporting

Field	Type	Description
Visit ID	Auto-generated	Auto-generated unique reference.
Inmate ID	Foreign Key	
Doctor	Foreign Key	
Date & Time	Time	Exact timestamp of the incident.
Diagnosis	String	Description

8. Machine Learning Features

CPMS integrates three AI/ML prediction modules to support proactive prison administration.

8.1 Risk Behavior Prediction

Attribute	Detail
Purpose	Predict which inmates are likely to engage in violent or disruptive behavior.
Input Signals	Incident history, disciplinary records, sentence type, block environment, social visit frequency.
Output	Risk score (Low / Medium / High / Critical) with contributing factors.
Consumers	Super Admin dashboard, Prison Manager dashboard.
Update Frequency	Daily re-scoring for all active inmates.

8.2 Overcrowding Prediction

Attribute	Detail
Purpose	Forecast which facilities are likely to exceed capacity within a defined time window.
Input Signals	Current occupancy, upcoming releases, pending admissions, transfer trends, sentence durations.
Output	Per-prison occupancy forecast for the next 30, 60, and 90 days.
Consumers	Super Admin dashboard (system-wide view), Prison Manager (facility-specific).
Alert Threshold	Alert triggered when projected occupancy exceeds 90% of capacity.

8.3 Recidivism Risk Scoring

Attribute	Detail
Purpose	Estimate the likelihood that an inmate will re-offend after release.
Input Signals	Age, offense type, sentence duration, education level, visit frequency, vocational training participation, prior offenses.
Output	Recidivism score (0–100) and recommended rehabilitation interventions.
Consumers	Super Admin, Prison Manager, Social Workers.

Use Case	Prioritize rehabilitation resources and inform parole recommendations.
----------	--

9. Dashboard Specifications

9.1 Super Admin Dashboard

Widget	Description
Total Prisoners per Prison	Bar chart showing inmate count for each facility.
Occupancy Rate	Gauge or percentage indicator per prison; color-coded (green < 75%, amber 75–90%, red > 90%).
Transfer Statistics	Counts of pending, approved, and denied transfer requests in the current period.
System Alerts	List of ML-generated flags: overcrowding risk, high-risk inmates, upcoming mass releases.

9.2 Prison Manager Dashboard

Widget	Description
Occupancy Rate	Block-by-block and cell-by-cell occupancy heatmap.
Active Incidents	Count of unresolved incidents in the facility, with links to reports.
Visit Requests	Queue of pending visit requests awaiting approval.
Upcoming Releases	List of inmates due for release within the next 30 days.
Pending Transfers	Outbound transfer requests awaiting Super Admin approval.

9.3 Officer Dashboard

Widget	Description
Block Occupancy	Cell-by-cell occupancy view for the officer's assigned block(s).
Recent Incidents	Incidents reported in the block within the past 7 days.
Today's Movements	Scheduled court appearances, hospital visits, or external movements for block inmates.
Solitary Confinement Records	Active solitary confinement orders and their expiry dates.

10. Non-Functional Requirements

10.1 Security

- All data transmission must use TLS 1.2 or higher.
- Role-Based Access Control (RBAC) enforced at both API and UI layer.
- All user actions on inmate data must be logged in an immutable audit trail.
- Disciplinary and legal visit records must be tamper-evident.
- Session timeouts for idle users: 15 minutes for officer roles, 30 minutes for manager roles.

10.2 Performance

- Dashboard pages must load within 3 seconds under normal load.
- Inmate search results must return within 1 second for queries on up to 100,000 records.
- ML model inference (risk scoring) must complete within 5 seconds per inmate.

10.3 Availability

- Target uptime: 99.9% (excluding planned maintenance windows).
- Planned maintenance must be scheduled during off-peak hours with advance notice.

10.4 Scalability

- The system must support up to 500 concurrent users across all roles.
- Must support adding new prisons without architectural changes.

10.5 Compliance

- Must comply with national data protection regulations for sensitive personal data.
- Legal visit records must maintain attorney-client privilege — inaccessible to officers.
- Solitary confinement records must enforce the 30-day maximum as a system constraint.

11. Glossary

Term	Definition
CPMS	Centralized Prison Management System — the product described in this PRD.
Super Admin	The national-level system administrator who oversees all prisons.
Prison Manager	The administrator responsible for a single prison facility.
Officer	A security guard assigned to one or more blocks within a prison.
Block	A named wing or section within a prison (e.g., Block A).
Cell	An individual room within a block that houses one or more inmates.
Transfer Request	A formal request to move an inmate from one prison to another.
RBAC	Role-Based Access Control — a method of restricting access based on user role.
Recidivism	The tendency of a released inmate to re-offend and return to incarceration.
Solitary Confinement	Administrative segregation of an inmate from the general population.
Legal Visit	A confidential consultation between an inmate and their legal counsel.
ML	Machine Learning — AI techniques used for predictive analytics in CPMS.

End of Document — CPMS PRD v1.0